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Beyond the traditional: Extending academic libraries' roles in research integrity based on the causes of research misconduct

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ABSTRACT

This paper reports on an ongoing research project aiming to extend the roles of academic libraries in research integrity (RI) based on reexploring the causes of research misconduct (RM) and restructuring the relationships among them. Since the causes of RM have been thoroughly investigated, a systematic review approach was approved valid and adopted in this study. A total of 60 articles were selected and analyzed using a thematic analysis approach. The analysis pointed to 21 RM causes under four themes: individual, organizational, professional, and cultural causes. Three critical causes stand as the primary connectors among various causes: insufficient RI education and training, lack of explicit rules and regulations, and "publish or perish" pressure. On this basis, three key roles of academic libraries in RI services can be extended: RI educators and counselors, RI policy-making advisors, as well as evaluators of academic outcomes. As one of the earliest research projects that investigate the roles of academic libraries in RI based on RM causes, its findings can provide useful implications for formulating specific RI services in academic libraries.

Introduction

Research integrity (RI) is of paramount importance not only to individuals and institutions but also to national and global interests (Anderson, 2014; Roy & Edwards, 2023; L. Tang, 2019). It should be highlighted that ensuring RI stands as an important mission and responsibility of the library and information science (LIS) field (Gibson & O'Hanlon, 2020; Herr, 2019; Seadle, 2017). The International Federation of Library Associations and Institutions regards RI as one aspect of academic library support for research ethics (IFLA, 2022). RI abilities, such as proper citation practices, are integral to information literacy, which is a critical competency that academic librarians have been imparting to researchers and students over the past decades (ACRL, 2015).

As breaches of RI professional standards (Steneck, 2006), issues of research misconduct (RM) are not solely ethical dilemmas but also information-related challenges (Bishop et al., 2021; Bresnahan & Condon, 2023; Gibson & O'Hanlon, 2020; Herr, 2019; Seadle, 2017; Szomszor, 2020). The majority of RM instances fundamentally involve improper acquisition, processing, dissemination, and utilization of information. As experts in providing, organizing, locating, and evaluating information (Bronshteyn & Baladad, 2006; Wood, 2004) academic librarians should acknowledge that RM constitutes "the library's problem" (Greer et al., 2012, p. 251) and promote researchers' responsible use of information (Ard & Ard, 2019).

To address RM issues, academic librarians have traditionally developed their "teaching and consulting roles" to support RI in the context of information literacy (Caravello, 2008, p. 142). Further, they have continued to amplify their roles in RI through being as data services librarians (Bishop et al., 2021), RM behavior detectors (Seadle, 2017), and other characters. Concurrently, academic librarians have strengthened their collaboration with other units within research institutions (Benjes-Small et al., 2008; Wood, 2004). Despite the considerable efforts placed by academic libraries on RI, institutional RI policies often overlook the library's role as a supporting entity (Alenzuela et al., 2017). Academic libraries mostly act as "passive actors" in supporting RI (Seadle, 2017, p. 113), and their educating role in RI is largely limited to addressing plagiarism (Coates, 2014).

To adapt the roles of academic libraries to the current academic landscape, this research focuses on RM causes. Nowadays, the evolving and uncertain academic landscape, particularly with the emergence of artificial intelligence technologies like ChatGPT in recent years, requires

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academic libraries to environmentally scan causes affecting RM and extend their roles in RI (Auer & Krupar, 2001; Benjes-Small et al., 2008; Caravello, 2008). The complexity of RM issues also prompts academic librarians to delve deeper into the causes beyond superficial observations to develop services aligned with RI principles (Wood, 2004). Moreover, the implementation strategies for RM-preventing would remain ineffective without effectively understanding RM causes (Holtfreter et al., 2020) and their relationships (Mardani et al., 2020). Thus, identifying RM causes and structuring the relationships among them can be very useful to serve as evidence for academic libraries to extend their roles in RI and develop effective RI services.

As reported in the existing literature, RM causes can range from individual personality traits and organizational characteristics to systemic factors (Fanelli et al., 2022; Franzen et al., 2007; Horbach et al., 2019). However, the existing findings are rather fragmented. Not only is there an evident lack of systematic efforts in generating a holistic understanding of the causes, but it is still unclear and urgently requires further exploration of how the causes of RM can influence research practices (Aubert Bonn & Pinxten, 2019; Fanelli et al., 2017, 2022; Gopalakrishna et al., 2022; Holtfreter et al., 2020; Vučković et al., 2020). Moreover, different levels of RM causes are "often held separate" since the relationship among diverse RM causes has not been completely established (Horbach et al., 2019, p. 416).

To address these gaps, this research adopts a bottom-up approach to reinforce the roles of academic libraries in RI instead of a top-down approach (articulating library services in accordance with the current RI policies and regulations). The starting point of this research is to identify RM causes and then restructure the relationships among them. This study reports on a systematic review, which is a critical research procedure that provides a theoretical basis for the subsequent research stages. To our knowledge, this is one of the earliest research projects that investigate the roles and participation of academic libraries in RI based on RM causes. Therefore, the following research questions are established:

RQ1. What are the causes of RM?
RQ2. What are the relationships among the causes of RM?
RQ3. What roles should academic libraries play based on RM causes?

Literature review

Research misconduct

For decades, RM has always been a controversial topic in the international scientific community (Aubert Bonn & Pinxten, 2019; Kaiser, 1999). There is no standard, universally accepted definition of RM (Bornmann, 2013). The most widely used RM definition, provided by the U.S. Office of Science and Technology Policy (OSTP, 2000), refers to "fabrication, falsification, or plagiarism in proposing, performing, or reviewing research, or in reporting research results". In this definition, fabrication and falsification "distort the creation of scientific knowledge", while plagiarism "fails to connect the knowledge to its origin" (Haven & van Woudenberg, 2021, p. 544). The OSTP definition does not include honest error or differences of opinion. It is widely adopted as the base definition in many publications on RM (Bornmann, 2013), and by scientific institutions when formulating their own RM frameworks (e.g., All European Academies, 2023; National Natural Science Foundation of China, 2022; The UK Research Integrity Office, 2023).

Although some scholars have tried to redefine and extend the RM definition from the simple three deadly sins (e.g., Fanelli, 2013), academia seems to prefer the minimalist OSTP definition (Kaiser, 1999). RM is regarded as the worst research behavior while responsible conduct of research (RCR) is considered as ideal behavior at another extreme. Falling someplace between them, questionable research practices (QRPs), such as misrepresentation, inaccuracy, and bias, are troubling but not serious enough (Steneck, 2006) so fall below the threshold of RM (Xie et al., 2021). However, some researchers still do not distinguish QRPs from RM clearly or regard QRPs as a kind of RM (e.g., Maggio et al., 2019; Tang, 2023).

As the international scientific community has dramatically become competitive (Fang & Casadevall, 2015; Reif, 1961; Rodrigues et al., 2023), there is an increasing number of RM cases compromising RI (Golden et al., 2023; Martinson et al., 2005). RM leads to severe consequences not just for researchers themselves, but also for the universities, funding agencies, and research participants (UKRIO, 2023). RM behaviors have significantly undermined the credibility and public trust in the scientific community, as well as severely compromised scientists' motivation and morale to engage in innovative and explorative endeavors (Martinson et al., 2005; Nelson & Lubchenco, 2022).

Causes of research misconduct

The RM causes are complex, characterized by "secrecy and complexity" in RM research (DuBois et al., 2013, p. 321). A thorough comprehension of the underlying RM causes can help predict potential occurrences of RM, design effective interventions, inform policy decisions, and reinforce educational initiatives (Fanelli et al., 2015; Farahat, 2022; Horbach et al., 2020; Mardani et al., 2020). Understanding the causes of RM can serve as the first step for academic libraries to fulfill their role in RI (Herr, 2019).

Sovacool's (2008) tripartite narrative perspective suggests that RM causes can be explained from individual impurity, institutional failure, and structural crisis. Several studies on RM causes (e.g., Fanelli et al., 2022; Fierz et al., 2014; Horbach et al., 2019) largely support Sovacool's (2008) perspective. Some scholars correspond these three narrative modes to micro, meso, and macro levels respectively, informing that the cause of RM is a multilevel problem (Faria, 2015; Fierz et al., 2014; Mardani et al., 2020). Individual narrative of RM causes suggests that "there are scoundrels among scientists" (Sovacool, 2008, p. 276), indicating the existence of potential inherent deviance in wrongdoers' mindset (Fanelli et al., 2022; Franzen et al., 2007). This has been well explained by the "bad apple" theory (e.g., Haven & van Woudenberg, 2021; Olesen et al., 2018; Resnik, 2010). Institutional narrative highlights organizational causes deriving from the policies, procedures, and training within academic institutions (Were et al., 2023). Structural narrative, focusing on the values promoted by modern science (Sovacool, 2008), reveals that the pervasive emphasis on scholarly productivity is eroding the academic system (Davis et al., 2007). The phenomenon, known as "publish and perish", is identified as "the library's problem" (The Association of Research Libraries et al., 1998).

Besides the above three typologies of RM causes, Davis (2003) develops a model of RM causes (Fig. 1) supplementing cultural and situational causes. Situational causes refer to temporary and stressful circumstances that can induce scientists to engage in RM (Committee on Responsible Science et al., 2017b). These causes are extrinsic to individuals and can encompass a diverse array of situational challenges. Davis (2003) introduced culture into the research landscape of RM causes from Sellin's (1938) culture conflict theory, which indicated that scientists may engage in RM in some ways that they may consider justified but are not acceptable in other cultures.

As depicted in Fig. 1, each circle represents a dimension of RM causes. It is evident that there is "an overlap" among different dimensions (Davis, 2003, p. 191), implying that one cause may be associated with more than one dimension. Davis's (2003) model has been developed by numerous studies from diverse disciplines (e.g., DuBois et al., 2013; Fanelli et al., 2015; Petousi & Sifaki, 2020). Thus, it can be adopted as the preliminary framework for the purpose of this study.

The roles of academic libraries in research integrity

Many interest groups like government agencies, publishers and editors, academic associations, and research institutions have participated

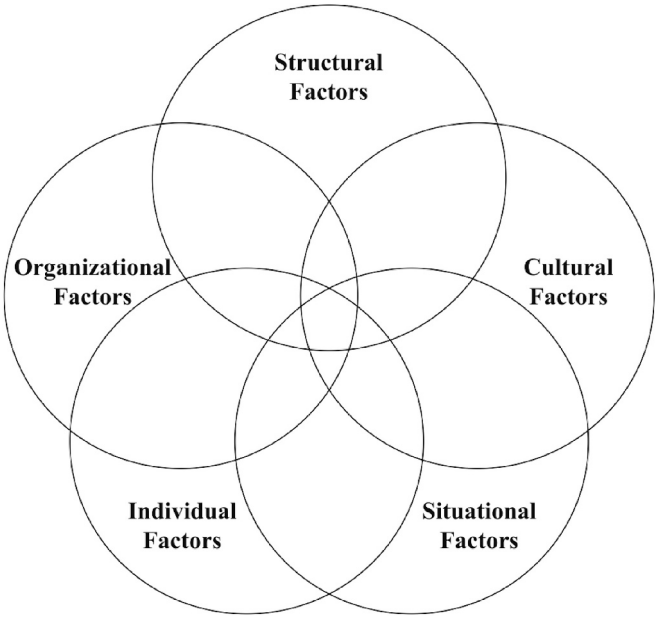


Fig. 1. The model of causes affecting RM proposed by Davis (2003).

in preventing, detecting, correcting, and reporting RM (Buljan et al., 2018; Gillespie, 1991; Gunsalus, 2019), as well as cultivating and repairing RI (Mejlgaard et al., 2020; L. Tang, 2019; Titus et al., 2008). Academic libraries have a shared responsibility with other groups and play a unique role in ensuring RI on the whole research lifecycle (Alenzuela et al., 2017; Borrego et al., 2018; Bresnahan & Condon, 2023; Seadle, 2017; Szomszor, 2020; Zhou et al., 2018). This is not merely because academic librarians can address sensitive RM issues with faculty and students without posing a threat, as they lack the authority to evaluate faculty and student work (Caravello, 2008). Furthermore, their long-standing expertise related to topics like scholarly publishing and intellectual property makes them highly attuned to RM matters (Caravello, 2008).

Academic librarians play a significant role in aiding faculty in detecting plagiarism in research papers and student work (Benjes-Small et al., 2008; Sharkey & Culp, 2005), as well as promoting the utilization of plagiarism detection tools (Alenzuela et al., 2017). Furthermore, academic libraries are sometimes called upon to assist in plagiarism detection for retracted articles in journals (Seadle, 2017). Administering punishment upon RM after its occurrence is justifiable, yet it is far less effective than preventing it proactively beforehand (Greer et al., 2012; Gunton, 2022; Wood, 2004). Academic librarians are proactive in educating faculty and students on how to prevent plagiarism by emphasizing the importance of correctly citing, critically thinking, and reasonably paraphrasing (Alenzuela et al., 2017; Auer & Krupar, 2001; Benjes-Small et al., 2008; Bronshteyn & Baladad, 2006; Caravello, 2008; Sharkey & Culp, 2005; Wood, 2004). Some librarians are also invited by the faculties to deliver in-class RI instruction to students (Alenzuela et al., 2017) or help integrate RI knowledge into the curriculum and assignments (Auer & Krupar, 2001; Wood, 2004).

Additionally, academic libraries have served as counselors to offer personalized RM counseling to faculty and students (Alenzuela et al., 2017; Auer & Krupar, 2001; Bronshteyn & Baladad, 2006; Caravello, 2008). Professional librarians are equipped to assist researchers in navigating conflicts of interest (COI) as they engage in scholarly publications (Gibson & O'Hanlon, 2020). Concurrently, academic libraries often organize various workshops to encourage researchers and students to engage in open discussions on RM (Alenzuela et al., 2017; Caravello, 2008; Wood, 2004). Specifically, as many disciplines increasingly become data-driven, research data management (RDM) has emerged as "fundamental to research integrity" (Committee on Responsible Science et al., 2017a, p. 168). RDM services gradually serve as a critical connection between academic libraries and RI (Herr, 2019). By offering RDM services, academic libraries have long been working closely with researchers (Bishop et al., 2021; Coates, 2014; Cox & Verbaan, 2016; Seadle, 2017).

The academic library is not "an island unto itself" (Greer et al., 2012, p. 252) from other units in supporting RI within research institutions. Librarians have established significant partnerships with various entities (Benjes-Small et al., 2008; Wood, 2004) such as writing centers (Ard & Ard, 2019), research integrity offices (Bishop et al., 2021), and internal departments responsible for COI (Gibson & O'Hanlon, 2020), yielding a greater impact than individual endeavors (Caravello, 2008). In some institutions, academic librarians are appointed to RI committees or similar divisions and even have the authority to make decisions on RM cases (Ard & Ard, 2019; Bresnahan & Condon, 2023; Wood, 2004).

Nonetheless, given the substantial changes in research practices today, heavily driven by artificial intelligence, the Internet, and digital resources, the traditional roles of academic libraries in RI are not completely feasible. Thus, it is strongly suggested to extend academic libraries' roles in RI with the RM cause-oriented map to navigate the complicated research landscape.

Research design and processes

Through a general review of existing literature, the research team could enhance their theoretical sensitivity, which according to Strauss and Corbin (1998), is critical for the analysis and theorization of qualitative data. The preliminary review not only confirms the research questions and objectives of this project but also shows that a systematic review can be very useful in responding to the research questions. Systematic review is a very structured, organized, and concentrated literature review strategy that "systematically search for, appraise and synthesis research evidence" (Grant & Booth, 2009, p. 102). Depending on different research aims and questions, systematic reviews can be either quantitative or qualitative (Hong et al., 2017; Hong & Pluye, 2018). Quantitative reviews are useful when validating and testing a theory, whereas qualitative reviews are more appropriate for exploring a phenomenon and theory development (Guetterman et al., 2015). Thus, a qualitative systematic review design was adopted in this study.

Whilst the preliminary review was designed to be general and wide, the research team intended to follow the systematic review procedures and perform a deep and rigorous analysis focusing on RM causes during the main review process. Three academic databases were used to collect relevant articles: Web of Science Core Collection, Scopus, and ProQuest Dissertations & Theses Global Full Text (PQDT). These databases jointly cover the core collection of peer-reviewed research articles and dissertations published across different disciplines. The retrieval strategy was set as:

> (cause OR "causal factor" OR contributor OR determinant OR driver OR factor OR incentive OR motivation OR reason) AND ((research OR academic OR scientific) AND (misconduct OR misbehavior) OR fabrication OR falsification OR plagiarism).

The retrieval strategy was designed to be inclusive of as many relevant articles as possible. To avoid unnecessary dilemmas in the RM definition, the retrieval strategy rigorously adopted the OSTP definition and excluded QRPs and dishonesty behaviors. The retrieval took place between May and June 2023, not restricting the publishing year of articles. The idea is to create a sufficient pool of articles, from which a manual search can be performed. The database retrieval returned a total of 1341 articles after removing duplications. Additionally, citation tracking was conducted manually, which allowed the identification of relevant articles not obtained from the database search. Using this approach, 21 new articles were identified.

Thereafter, the research team formulated a set of inclusion criteria to

make sure all included articles are of high relevance to this research. For inclusion, a study must meet the following criteria: (a) it was published in English; (b) it is one of the journal articles, conference proceeding articles, chapters, or dissertations; (c) it has been peer-reviewed before publication; (d) its full text is available for download; (e) it must report RM causes; (f) it must conform the OSTP definition of RM. Notably, the research team did not directly exclude an article during the screening process due to the presence of QRPs or dishonesty in the title, keywords, or abstract. Instead, the research team compared the OSTP definition with the research behaviors articulated in the full text to determine whether the article should be included. The article screening procedures are demonstrated in the PRISMA diagram (Fig. 2) as suggested by Page et al. (2021).

A total of 60 articles were selected eventually and uploaded to a qualitative data analysis software, MAXQDA. A thematic analysis was employed for the data analysis. Thematic analysis is one of the most foundational and widely used methods for all qualitative analysis (King et al., 2018; Zhou et al., 2020). It is suitable for this systematic review because this approach is particularly useful when a research project aims to discover themes and concepts embedded throughout qualitative data (Rubin & Rubin, 2011).

In this systematic review, the thematic analysis was operationalized as a systematic procedure of coding and representing research articles (Zhou et al., 2020). In the analysis, the articles, adopted as qualitative data, were analyzed with three types of coding techniques: open, axial, and selective coding, as suggested by Strauss and Corbin (1998). Using these coding techniques, the research team was able to go through the articles and systematically identify, interpret, and label RM causes in data. Moreover, codes were represented through a theoretical narrative, which describes and discusses all RM causes.

Review findings

The literature analysis pointed to 21 RM causes in four main themes: individual causes, organizational causes, professional causes, and cultural causes. The final classification of each RM cause is based on the frequency with which the included articles support that cause in each theme. All included articles reported at least one type of RM cause. The breakdown of RM causes organized by included articles is presented in Table 1.

Individual causes

The literature analysis reveals that certain individual causes like personality traits, ethical awareness, and economic desperation can eminently influence the occurrence of RM. The analysis pointed to six individual causes:

- Character flaws;
- Lack of ethical awareness;
- Lack of experience;
- Bad role model learning;
- Unstable mental state;
- Excessive personal and family pressures.

Several researchers argue that certain character flaws (e.g., narcissism, Machiavellianism, greed, laziness, and psychopathic traits) can result in RM (Aubert Bonn & Pinxten, 2019; Cairns et al., 2021; Rodrigues et al., 2023; Tijdink et al., 2016). Narcissism is the most frequently cited and discussed character flaw in reviewed articles. It denotes the personality characterized by the pursuit of self-gratification through vanity, arrogance, ego, or self-interest, and the quest for recognition of one's attributes (Buljan et al., 2018; Cairns et al., 2021; Tijdink et al., 2016). This trait can lead scientists to become excessively self-centered, reject to accept criticism or differing opinions, and even sabotage the work and data of others (Buljan et al., 2018; Cairns et al., 2021; Gillespie, 1991; Tijdink et al., 2016). Some scientists may even develop a Messianic complex, characterized by a strong desire to promote a

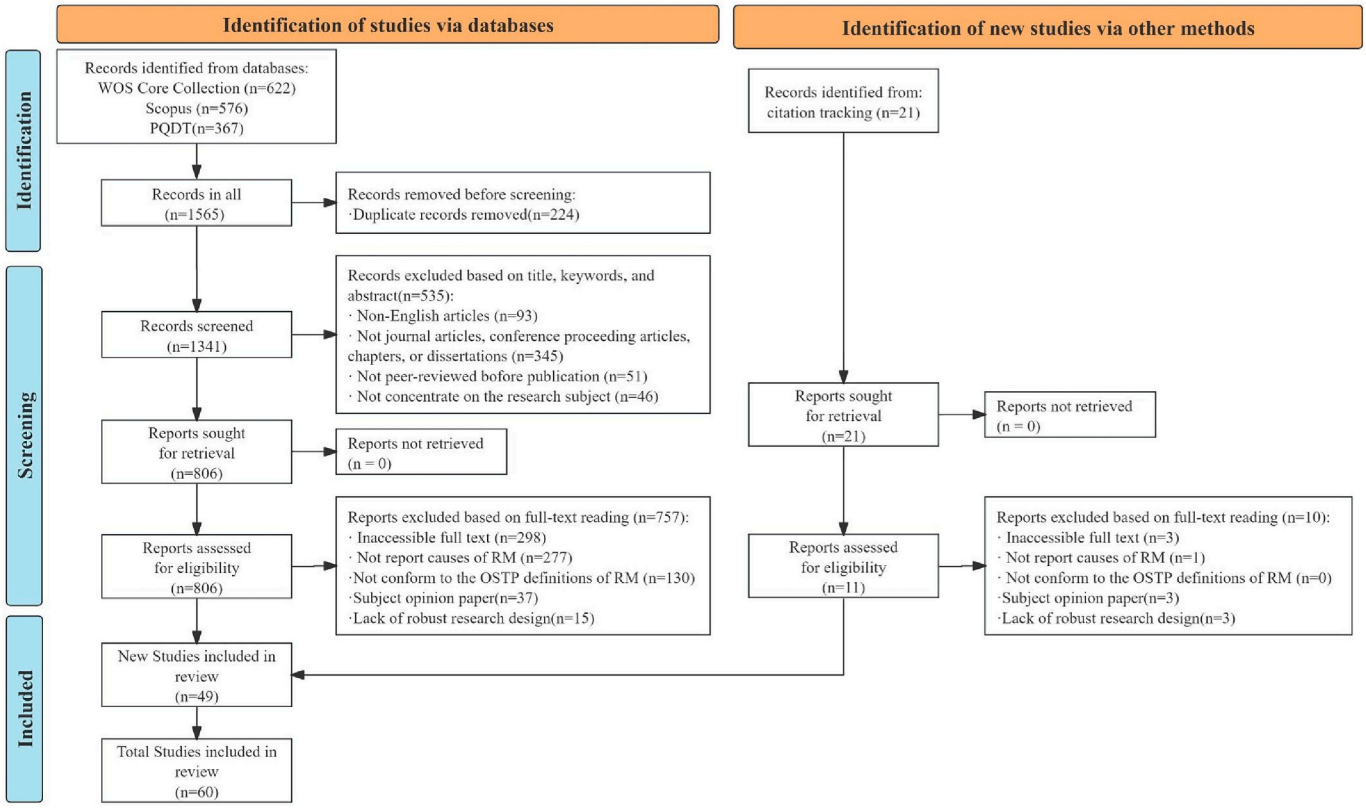


Fig. 2. PRISMA diagram on article screening processes.

Table 1
Mapping of RM causes from including studies.

No.	Literature	Individual Causes	Organizational Causes	Professional Causes	Cultural Causes
1	(Al-Has et al., 2021)	√	√	√	
2	(Anderson et al., 2007)			√	
3	(Aubert Bonn & Pinxten, 2019)	√	√		√
4	(Aubert Bonn & Pinxten, 2021)	√	√	√	√
5	(Buljan et al., 2018)	√	√	√	√
6	(Cairns et al., 2021)	√	√	√	
7	(Casadevall & Fang, 2012)		√	√	√
8	(Choupani et al., 2018)	√			
9	(Committee on Responsible Science et al., 2017b)	√	√	√	√
10	(Craig et al., 2021)	√	√		
11	(Dannhoferová et al., 2022)	√	√		√
12	(Davis, 2003)				√
13	(Davis et al., 2007)	√	√	√	√
14	(DuBois et al., 2013)	√		√	
15	(ElHafeez et al., 2022)	√			
16	(Fanelli et al., 2017)		√	√	√
17	(Fanelli et al., 2019)	√	√	√	√
18	(Fanelli et al., 2015)	√			
19	(Fanelli et al., 2022)	√	√	√	√
20	(Fang et al., 2013)	√	√	√	√
21	(Farahat, 2022)	√	√	√	√
22	(Faria, 2015)		√	√	
23	(Fierz et al., 2014)	√	√	√	
24	(Fisher & Partin, 2014)		√		
25	(Franzen et al., 2007)	√	√	√	
26	(Fonseca et al., 2016)	√	√	√	√
27	(Gillespie, 1991)	√	√	√	√
28	(Golden et al., 2023)	√	√	√	
29	(Gopalakrishna et al., 2022)		√	√	√
30	(Haven & van Woudenberg, 2021)	√	√	√	
31	(Herr, 2019)	√	√	√	
32	(Holtfreter et al., 2020)	√	√	√	√
33	(Horbach & Halffman, 2019)	√	√	√	
34	(Horbach et al., 2019)	√	√	√	
35	(Horbach et al., 2020)		√		√
36	(Juyal et al., 2015)	√	√		
37	(Krstić, 2015)	√	√		√
38	(Maggio et al., 2019)	√	√	√	√
39	(Makarova, 2019)	√	√		√
40	(Mardani et al., 2020)	√	√	√	
41	(McCabe et al., 2008)	√			√
42	(Mirshekary & Lawrence, 2009)				√
43	(Okonta & Rossouw, 2013)	√	√	√	
44	(Olesen et al., 2017)				√
45	(Olesen et al., 2018)	√	√	√	√
46	(Parnther, 2022)	√			√
47	(Patnaik, 2016)		√	√	
48	(Putra et al., 2023)				√
49	(Rodrigues et al., 2023)	√	√	√	√
50	(Roje et al., 2022)	√	√	√	√
51	(Roy & Edwards, 2023)	√	√	√	
52	(Sadeghi-Bazargani et al., 2020)	√	√	√	
53	(Schulz, 2015)		√		
54	(Tijdink et al., 2016)	√		√	
55	(Tourish & Craig, 2020)		√		
56	(Vučković et al., 2020)	√	√		
57	(Williams et al., 2014)				√
58	(Wright et al., 2008)	√	√	√	
59	(Yeo-Teh & Tang, 2022)			√	
60	(Yu et al., 2021)	√	√	√	√
Total (n)		44	46	39	31
Total (%)		73.33	76.67	65.00	51.67

particular theory or prove their hypotheses at the expense of RM (Davis et al., 2007).

Moreover, a deficiency in ethical awareness stands as a primary underlying cause of RM (Cairns et al., 2021; Rodrigues et al., 2023). This factor manifests in individuals' attitudes toward RM and their perceptions of RI (ElHafeez et al., 2022), leading them to disregard the consequences of RM (Buljan et al., 2018; Holtfreter et al., 2020). In the absence of sufficient ethical awareness, scientists may rationalize and normalize RM within academic evaluations (Aubert Bonn & Pinxten, 2019; Wright et al., 2008).

It can be challenging for early-career, and thus inexperienced researchers to discern acceptable practices from unacceptable ones (Buljan et al., 2018; Horbach et al., 2020; Horbach & Halffman, 2019; Okonta & Rossouw, 2013; Roje et al., 2022). In such circumstances, they may be inclined to imitate their mentors and supervisors (Roje et al., 2022) or be overly influenced by their peers (Parnther, 2022). If their role models engage in RM and are not held accountable, there is a risk for them to replicate these behaviors (Buljan et al., 2018; Roje et al., 2022; Vučković et al., 2020).

Poor mental state has been linked to instances of RM (Cairns et al.,

2021; Davis et al., 2007; Vučković et al., 2020). Several studies suggest that fear of loss and making mistakes can significantly contribute to the stress experienced by scientists, potentially leading them to resort to RM (Cairns et al., 2021; Committee on Responsible Science et al., 2017b; Parnther, 2022). The "subject norms" of scientists to satisfy expectations exacerbate social pressures on their mental strain (ElHafeez et al., 2022, p. 14). Notably, excessive personal and family pressures, such as financial difficulties (Roje et al., 2022; Sadeghi-Bazargani et al., 2020), the challenges faced by female scientists related to childbirth (Holtfreter et al., 2020), and the academic performance expectations placed on research students by their parents (Farahat, 2022; Vučković et al., 2020), are significant contributors to emotional instability and heighten the likelihood of RM.

Organizational causes

The analysis shows that a research institution with a strong RI culture can effectively prevent RM from occurring (Horbach et al., 2019). However, the responsibility of dealing with RM is still unclear due to the ambiguous roles of different stakeholders, who can easily evade responsibilities and "pass the buck" to others (Craig et al., 2021, p. 298). Some organizations may delegate RM responsibilities to individuals (Horbach et al., 2019), whereas several organizations deliberately "abdicate responsibilities for ensuring the integrity of the research productivity" (Golden et al., 2023, p. 76). In the literature analysis, organizational causes place more emphasis on the responsibility of academic institutions. Specifically, six organizational causes were identified:

- Insufficient RI education and training;
- Lack of explicit rules and regulations;
- Lack of report and transparency on RM;
- Lack of punishment and insufficient censure;
- Lack of supervision and mentoring;
- Misdirected reward system.

Despite the long history of RI education and training in many organizations, it has been criticized as ineffective (Cairns et al., 2021; Mardani et al., 2020), inadequate (Fisher & Partin, 2014), inappropriate (Fonseca et al., 2016), or overly general in content (Roje et al., 2022). More seriously, not all researchers have the opportunity to receive RI training (Krstić, 2015). Some important skills, such as data processing (Al-Has et al., 2021), ethical scientific writing (Juyal et al., 2015; Rodrigues et al., 2023), and proper reporting (Sadeghi-Bazargani et al., 2020), may not be instilled in researchers through RI training, making them lack of basic knowledge on RI (Aubert Bonn & Pinxten, 2021; Rodrigues et al., 2023; Roje et al., 2022). It is particularly pronounced that researchers from non-English-speaking nations are at a higher risk of inadvertently falling into the plagiarism trap when publishing their work in English (Al-Has et al., 2021; Haven & van Woudenberg, 2021; Juyal et al., 2015). Moreover, it is crucial to incorporate correct citation skills into RI training, as researchers who do not learn to cite properly may unintentionally commit plagiarism (Fisher & Partin, 2014).

Furthermore, RM proliferates in the absence of "strong institutional safeguards" (i.e. policies, regulations, procedures, rules, guidelines, etc.) (Tourish & Craig, 2020, p. 3). Although some organizations have established regulations to address RM behavior, the majority remain outside the regulatory purview (Golden et al., 2023; Rodrigues et al., 2023). The ambiguities (Horbach & Halffman, 2019; Juyal et al., 2015) and ineffectiveness (Okonta & Rossouw, 2013; Sadeghi-Bazargani et al., 2020) inherent in these existing regulations create opportunities for RM practices to persist. For example, the lack of uniformity in the definition and manifestation of RM serves as a significant loophole for wrongdoers to evade accountability (Franzen et al., 2007; Golden et al., 2023; Horbach & Halffman, 2019; Mardani et al., 2020; Okonta & Rossouw, 2013; Rodrigues et al., 2023).

The lack of related regulations in research organizations means that investigating and reporting RM is not mandatory for them (Golden et al., 2023). Together with the high necessity of time, effort, cost, and expertise to do this work (Juyal et al., 2015; Olesen et al., 2018), some organizations are reluctant to investigate and report RM for fear of damage to their reputation or other adverse effects (Roje et al., 2022). More problematically, to safeguard their reputation, certain organizations even forbid their employees from providing evidence and clues during RM investigations (Schulz, 2015), openly discussing RM incidents (Gopalakrishna et al., 2022), and engaging in peer criticism (Buljan et al., 2018; Franzen et al., 2007). The lack of transparency regarding RM drives individuals to engage in RM more frequently, as their actions will go unreported and unpunished (Farahat, 2022).

Sometimes, even though RM cases have been reported, there is a discrepancy between the actual punishment and the reported explanation (Horbach et al., 2019). A deficient punishment system and insufficient censure contribute to the prevalence of RM (Buljan et al., 2018; Cairns et al., 2021; Holtfreter et al., 2020; Okonta & Rossouw, 2013). Organizational procedures of RM punishment can be complex, often resulting in milder results (Horbach et al., 2019) than the widespread public expectations of severe punishment (Holtfreter et al., 2020).

Numerous studies (e.g., Al-Has et al., 2021; Gopalakrishna et al., 2022; Mardani et al., 2020; Yu et al., 2021) posit that insufficient supervision and mentoring of organizations during the research process can lead to RM. Such dereliction of supervising and mentoring responsibilities can manifest in three relationships: from organizations to researchers (ElHafeez et al., 2022), from senior researchers to junior researchers (Gillespie, 1991), and from mentors to their mentees (Olesen et al., 2018). This problem can be attributed to "the […] cover up by the organizations" for RM (Patnaik, 2016, p. 1247), lack of training for mentoring staff (Juyal et al., 2015), understaffing for supervision (Herr, 2019), etc.

Several research organizations have adopted the "winner-takes-all" reward systems (Casadevall & Fang, 2012, p. 892). Under this misdirected system, scientists are more prone to astray given the high potential for reward through RM and the low likelihood of getting punished (Herr, 2019). Particularly, cash incentives tied to individual publication performance are quite risky (Aubert Bonn & Pinxten, 2019; Fanelli et al., 2015, 2017; Vučković et al., 2020), which may not directly lead to RM, but can reflect the values of research organizations that might encourage RM (Fanelli et al., 2019, 2022).

Professional causes

The literature analysis showed that certain RM causes are not very personal and can thrive across different organizations. The analysis identified four professional causes:

- "Publish or perish" pressure;
- Pressure from grant application;
- Pressure from promotion and tenure;
- Overly intense professional competition.

In scholarly discourse, the mantra "publish or perish" has become an increasingly pervasive ethos driving RM (Cairns et al., 2021; Golden et al., 2023; Holtfreter et al., 2020; Maggio et al., 2019; Patnaik, 2016; Tijdink et al., 2016; Yu et al., 2021). The pressure to publish can lead scientists to prioritize quantity over quality (Buljan et al., 2018), cut corners during the research (Franzen et al., 2007), pursue publishing in high-impact journals (Holtfreter et al., 2020), and reject retractions when necessary (Golden et al., 2023). These substandard practices precariously position scientists on the threshold of RM. The increase in publication volume due to publishing pressure intensifies the burden on journal editors and the peer review system, necessitating heightened alertness to maintain RI (Aubert Bonn & Pinxten, 2021; Casadevall & Fang, 2012).

The imperative of prolific publication is a critical determinant of career advancement and the application of research funding (Olesen et al., 2018). The evaluative criteria employed in grant assessment procedures underscore previous publications (Patnaik, 2016), thereby compelling scientists to maximize their publication count to align with funding agencies' expectations or secure additional financial support (Buljan et al., 2018; DuBois et al., 2013; Franzen et al., 2007; Holtfreter et al., 2020). Similarly to journal editors, funders also face significant pressure to manage a large volume of applications while simultaneously grappling with the challenge of recruiting qualified peer reviewers (Aubert Bonn & Pinxten, 2021).

The pursuit of tenure and promotional advancements place significant pressure on scientists to publish more (Davis et al., 2007; Golden et al., 2023; Okonta & Rossouw, 2013; Tijdink et al., 2016; Wright et al., 2008). Junior scholars, in particular, are acutely susceptible to this pressure, with their career continuity hinging on their publication output (Holtfreter et al., 2020). However, a sudden decrease in scholarly productivity can be observed in several scientists after achieving the tenure goal (Buljan et al., 2018). In this sense, publishing has become "a self-serving activity" deviating from the ethos of knowledge contribution (Buljan et al., 2018, p. 228).

The "publish or perish" pressure is usually amalgamated with the pressures from securing funding and achieving tenure (Golden et al., 2023; Herr, 2019; Okonta & Rossouw, 2013; Yu et al., 2021), forming "a self-propagating vicious cycle" (Patnaik, 2016, p. 1246). This interplay of pressures fosters a fiercely competitive professional environment within the scientific community, which is especially pronounced in fields surrounded by funding scarcities and limited tenure-track opportunities (Farahat, 2022; Franzen et al., 2007; Gopalakrishna et al., 2022; Herr, 2019; Rodrigues et al., 2023; Vučković et al., 2020). Such a hypercompetitive climate, in turn, inadvertently incubates the conditions conducive to RM (Golden et al., 2023).

Cultural causes

Cultural causes relate to the values and experiences that shape scientists' upbringing and professional environment (Olesen et al., 2018). The culture-level causes exhibit a stronger effect than individual-level causes on RM (Fanelli et al., 2019). Through the literature analysis, five cultural causes emerged:

- Cultural discrepancy and conflict;
- Strong authority hierarchy;
- Collectivist and Individualist culture;
- Gender differences;
- Low uncertainty tolerance.

Cultural discrepancy regarding the conceptual cognition and acceptance level of RM can result in its occurrence (Roje et al., 2022). For instance, some cultures may inadvertently encourage students to achieve goals through unethical practices (Buljan et al., 2018; Roje et al., 2022). In certain contexts, duplicating existing work may be misconceived as respect for the original, thus bypassing the label of plagiarism (Parnther, 2022). Influenced by these problematic cultural norms, scientists may exhibit neutralizing attitudes toward RM without any guilt (Farahat, 2022; Makarova, 2019). Foreign scientists, in particular, may struggle with RM due to the conflicting cultural perceptions of RM between their home and host countries (Davis, 2003; Olesen et al., 2017; Parnther, 2022). This cultural conflict may precipitate RM, as foreign scientists face unique pressures to reconcile the new, but more internationally recognized, academic norms with those of their home cultures (Davis et al., 2007).

Moreover, the risk of RM is higher in countries with more rigid hierarchies because of the precedence of respect for authority over mutual criticism in such societies (Aubert Bonn & Pinxten, 2021; Fanelli et al., 2015, 2017). Strong authority hierarchy tends to cultivate a culture of deference toward those in positions of authority such as principal investigators or senior researchers in academia (Yu et al., 2021). Since science is often a collaborative endeavor (Casadevall & Fang, 2012) and senior researchers control more resources (Horbach et al., 2020), junior researchers tend to rely on those with strong authority for opportunities and support (Faria, 2015). Consequently, even when junior researchers and doctoral students witness RM conducted by their senior counterparts, they might not choose to be the whistle-blowers (Buljan et al., 2018; Krstić, 2015). More seriously, power-holders in a strict hierarchical culture even have the privilege to dictate what is plagiarism, whether plagiarism belongs to the category of RM or not, as well as who should be punished due to RM (Putra et al., 2023).

The influence of collectivist culture on RM is substantial and multifaceted (Farahat, 2022; Mirshekary & Lawrence, 2009; Parnther, 2022). In such cultures, scientists prioritize loyalty to peers and institutions as more important than compliance with RI principles (Parnther, 2022) and are less inclined to take RM as serious violations compared to those from individualist cultures (Williams et al., 2014). In collectivist societies, the "culture of shame" for failing to meet social expectations can drive scientists to commit RM to reach their objectives (Fanelli et al., 2022, p. 14). This culture, predicated on cooperative values, can predict certain RM behaviors, especially during research collaboration (Makarova, 2019; McCabe et al., 2008). Nonetheless, individualistic cultures are not exempt from RM. The heightened emphasis on personal achievements within individualistic cultures may amplify the incidence of RM, as it reduces collective oversight and cooperation (Olesen et al., 2018).

Gender differences constitute another RM cause reflecting cultural contexts (Fang et al., 2013). Academia is a masculine society (Casadevall & Fang, 2012), a manifestation of which is evident in the predominance of male scientists implicated in RM cases (Fang et al., 2013; Farahat, 2022; Makarova, 2019). Compared with female scientists, psychological characteristics typically ascribed to male scientists, such as aggressiveness, competitiveness, and assertiveness, predispose them to seek status and take risks through RM (Fang et al., 2013). Additionally, male scientists are more likely to exhibit permissive and accepting attitudes toward RM (Rodrigues et al., 2023), in contrast to their female counterparts who are more unacceptable and afraid of committing RM (Mirshekary & Lawrence, 2009; Yu et al., 2021).

Cultural cause characterized by the low tolerance for uncertainty reveals a higher tendency for engagement in RM among scientists (McCabe et al., 2008; Mirshekary & Lawrence, 2009). For instance, job insecurities of science can be a trigger for RM in such cultural settings, as scientists tend to mitigate uncertainty through RM (Aubert Bonn & Pinxten, 2021; Davis et al., 2007). Educational practices that focus on memorization and rote learning can lead to plagiarism because they often produce scientists with a low tolerance for uncertainty, favoring stability and conformity over critical thinking and originality (Dannhoferová et al., 2022; Parnther, 2022; Rodrigues et al., 2023).

Discussion

RQ1: what are the causes of RM?

The thematic analysis of this systematic review identifies 21 RM causes under four main themes: individual causes, organizational causes, professional causes, and cultural causes. The research findings corroborate Sovacool's (2008) tripartite narratives regarding RM causes and align with part of Davis's (2003) model. Furthermore, the research team expands upon the cultural perspective outlined in Davis's (2003) model and replaces the structural and situational causes suggested by Davis (2003) with "professional causes", as the former two causes can be essentially traced back to professional competitions and pressures among scientists (Davis et al., 2007; Faria, 2015).

As "scholarly integrity can vary from culture to culture" (JAL, n.d.), cultural causes influencing RM may arise not only from cultural conflicts

as Davis (2003) suggested, but also from national cultural traits as revealed by four additional cultural causes (i.e. strong authority hierarchy, collectivist and individualist culture, gender differences, and low uncertainty tolerance). These causes align with high power distance, collectivism versus individualism, masculinity versus femininity, and high uncertainty avoidance respectively in Hofstede's theory of cultural dimensions. Thus, Hofstede's theory of cultural dimensions is useful to measure and explain the effect of different national cultures on RM (Zhou & Nunes, 2013). For instance, traditional masculinity norms emphasize competition, dominance, and the pursuit of recognition and prestige (Hofstede et al., 2010, pp. 137–138), fostering an environment where scientists are more inclined to achieve impressive results through RM.

There are several distinctions in the categorization of RM causes between this study and prior research. For example, this study views "insufficient RI training and education" as an organizational cause, while Cairns et al. (2021) tend to view it as an individual cause because the high-quality RI education provided by organizations can ultimately be rendered ineffective without the active engagement and serious attention of scientists. Thus, each RM cause may fall under different dimensions and can influence RM from multiple angles, supporting the overlapping relationships among various dimensions of RM causes in Davis's (2003) model.

RQ2: what are the relationships among the causes of RM?

The research findings demonstrate that RM causes are potentially interrelated and influential. Thus, the research team decided to revisit and reanalyze the selected articles to identify evidence to support underlying theme-theme, theme-code, and code-code relationships. For example, "lack of report and transparency on RM" can be partly attributed to "lack of explicit rules and regulations", because

> "institutional and systemic policies and accepted patterns of behavior that allow such misconduct to go undetected, unreported, or ignored, run the risk of calling into question the field's ability to self-police, self-correct, […]" (Golden et al., 2023, p. 80).

Through the conceptualization, a conceptual model was developed (Fig. 3).

As shown in Fig. 3, cultural causes are placed at the center of the conceptual model, suggesting that other dimensions of RM causes have cultural origins. The interconnections among individual, organizational, and professional causes can contribute to the "social origin" of RM (Horbach et al., 2019, p. 416). Professional causes are prevalent across different organizations, indicating a shared set of causes influencing most organizations. Although there is variability in how organizational causes manifest among various institutions, it implies that the context of each institution may affect the extent of these causes. This variability and prevalence suggest a reciprocal relationship between professional and organizational causes. Moreover, individual causes are subject to the effects of both organizational and professional causes since scientists can be influenced by both their affiliated organizations and the whole academic system.

Davis's (2003) model demonstrated that all dimensions of RM causes were equal and overlapping. This systematic review, however, reconstructs the relationship network by explicating the interactions not only between different dimensions but also among various RM causes. The new conceptual model reveals a more complex and hierarchical structure among RM causes. The model represents the first stage of theoretical development. Further research can explore the specific impact of each cause on RM and the relationship among different causes through empirical studies. Particularly, academic libraries should not be



Fig. 3. A conceptual model of the review findings.

absent from these further studies.

RQ3: what roles should academic libraries play based on RM causes?

In the conceptual model, three critical causes stand as the primary connectors among various causes: insufficient RI education and training, lack of explicit rules and regulations, and "publish or perish" pressure. When academic libraries leverage their expertise to develop RI services, they should prioritize these critical causes while coordinating with other causes. Thus, extending from the traditional roles of academic libraries in RI, this research proposes three key roles academic libraries should play in RI based on RM causes:

- *RI Educators and Counselors.* Firstly, academic libraries ought to adopt educational approaches to boost researchers' ethical awareness, enhance their academic writing abilities, and enrich their RI knowledge. Addressing user feedback regularly is vital for refining the existing education initiatives to ensure effectiveness and relevance. Secondly, teaching and consulting services offered by academic libraries should cater to researchers at various stages of their research careers, particularly to novice students and junior researchers. Staff handling RM incidents and RI educators should undergo specialized training, too. Thirdly, academic libraries should provide specialized teaching and consulting services to foreign researchers and students to help them understand the differences in RI guidelines between their home country and the host country, so as to prevent them from inadvertent violations of RI principles due to cultural discrepancies and conflicts.
- *RI Policy-Making Advisors.* Academic librarians can serve as advisors in formulating rigorous policies to compensate for the absence of explicit regulations, even if they typically lack the direct authority to enact RI policies. These policies must first clarify the roles of academic libraries and other units in RI, as well as provide precise definitions and descriptions of RM. They should mandate RI education as a preventive measure against RM. Moreover, academic librarians are obligated to assist institutions in elucidating the supervision process during scientific research, as well as the procedures for reporting RM cases and enforcing policy-based penalties. Overall, comprehensive regulations need to be established within organizations to address RM, covering the entire research lifecycle.
- *Evaluators of Academic Outcomes*. Academic libraries should advocate for a scientific evaluation framework that breaks the monopoly of publication quantity to alleviate the "publish or perish" pressure. They can promote the use of alternative and qualitative metrics for assessing research impact, rather than solely relying on quantitative measurements like journal impact factors. Likewise, providing tailored guidance to researchers on emphasizing quality over quantity is crucial. Libraries can also promote open science practices, such as unrestricted access to publications and research data, to amplify the transparency, visibility, and impact of academic outcomes. By relieving researchers of the constant publishing pressure, academic libraries can help mitigate the stress associated with research funding and tenure promotion.

While maintaining the traditional "teaching and consulting roles" of academic libraries (Caravello, 2008, p. 142), this research advocates for a deeper and more proactive role for libraries in RI education. It emphasizes the need to move beyond solely addressing plagiarism and to instead actively contribute to preventing RM causes. While some academic libraries have supported policy-making in the past, their role within the broader RI framework has not been fully recognized. This research proposes an expanded role for academic libraries as RI policy-making advisors, offering specific recommendations for addressing RM before, during, and after it occurs. Additionally, it introduces the concept of libraries serving as evaluators of academic outcomes, a role not previously explored. Engaging as unbiased stakeholders, libraries can further enhance RI by providing valuable evaluations based on the quality of academic outcomes.

Conclusion

This research reports on a systematic review aiming to extend the roles of academic libraries in RI based on RM causes. The analysis of 60 articles pointed to 21 RM causes under four themes, indicating three key roles of academic libraries in RI services. Given the ubiquitous nature of RM, the scientific community must collaborate and develop a comprehensive approach to address specific RM causes. As professional institutions in the field of LIS, academic libraries are obligated to first recognize the evolving nature of their roles and adapt to new situations. Further, academic libraries can take on the responsibility of supporting RI guided by the RM cause-oriented map in the changing landscape of RM.

This research has several implications for both theory and practice. Theoretically, this research provides a comprehensive scenario of RM causes and restructures a complex and hierarchical relationship framework among them. Thereby, it contributes to extant knowledge by shifting the focus of academic libraries in RI from traditional teaching and consulting to RM cause-oriented multiple services. Additionally, this research suggests further robust empirical studies to test on how these causes affect RM and the relationships among them whilst the findings have offered certain plausible explanations. Practically, this research extends traditional roles of academic libraries in RI to adapt them to today's academic landscape and provides tailored strategies to enable them play a larger role in supporting RI. Academic libraries can develop specific RI services based on the extended roles, or explore other roles suitable for their situations with the guidance oriented toward RM causes. Broadly, other interest groups in RI can utilize the research findings to articulate appropriate strategies in collaboration with academic libraries.

CRediT authorship contribution statement

Wei Feng: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Lihong Zhou:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Junmin Xiao:** Writing – review & editing, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

All authors have approved the manuscript and declared that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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